

1. A particle of mass 0.5 kg is moving with velocity $(2\mathbf{i} + 4\mathbf{j})\text{ m s}^{-1}$ when it receives an impulse of $(-4\mathbf{i} + 6\mathbf{j})\text{ N s}$.

(a) Find the speed of the particle immediately after it receives the impulse.

(5)

(b) Find the size of the angle between the direction of motion of the particle immediately before it receives the impulse and the direction of motion of the particle immediately after it receives the impulse.

(3)

2. A car of mass 600 kg tows a trailer of mass 200 kg up a hill along a straight road that is inclined at angle θ to the horizontal, where $\sin \theta = \frac{1}{20}$. The trailer is attached to the car by a light inextensible towbar. The resistance to the motion of the car from non-gravitational forces is modelled as a constant force of magnitude 150 N. The resistance to the motion of the trailer from non-gravitational forces is modelled as a constant force of magnitude 300 N.

When the engine of the car is working at a constant rate of P kW the car and the trailer have a constant speed of 15 m s^{-1}

- (a) Find the value of P .

(5)

Later, at the instant when the car and the trailer are travelling up the hill with a speed of 20 m s^{-1} , the towbar breaks. When the towbar breaks the trailer is at the point X . The trailer continues to travel up the hill before coming to instantaneous rest at the point Y . The resistance to the motion of the trailer from non-gravitational forces is again modelled as a constant force of magnitude 300 N.

- (b) Use the work-energy principle to find the distance XY .

(4)

3. A particle P of mass 0.25 kg is moving on a smooth horizontal surface under the action of a single force, $\mathbf{F}$ newtons.

At time t seconds ($t \geq 0$), the velocity $\mathbf{v}\text{ ms}^{-1}$ of P is given by

$$\mathbf{v} = (6 \sin 3t)\mathbf{i} + (1 + 2 \cos t)\mathbf{j}$$

- (a) Find $\mathbf{F}$ in terms of t .

(3)

At time $t = 0$, the position vector of P relative to a fixed point O is $(4\mathbf{i} - \sqrt{3}\mathbf{j})\text{ m}$.

- (b) Find the position vector of P relative to O when P is first moving parallel to the vector $\mathbf{i}$.

(6)

4. Two small balls, A and B , are moving in opposite directions along the same straight line on smooth horizontal ground. The mass of A is $2m$ and the mass of B is $3m$. The balls collide directly. Immediately before the collision, the speed of A is $2u$ and the speed of B is u . The coefficient of restitution between A and B is e , where $e > 0$

By modelling the balls as particles,

- (a) show that the speed of B immediately after the collision is $\frac{1}{5}u(1 + 6e)$. (6)

After the collision with ball A , ball B hits a smooth fixed vertical wall which is perpendicular to the direction of motion of B .

The coefficient of restitution between B and the wall is $\frac{5}{7}$

Ball B rebounds from the wall and there is a second direct collision between A and B .

- (b) Find the range of possible values of e . (4)
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5.

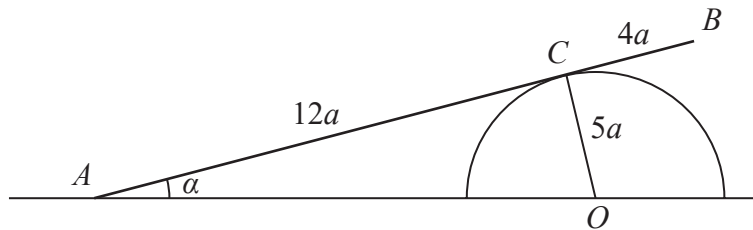


Figure 1

A smooth solid hemisphere is fixed with its flat surface in contact with rough horizontal ground. The hemisphere has centre O and radius $5a$.

A uniform rod AB , of length $16a$ and weight W , rests in equilibrium on the hemisphere with end A on the ground. The rod rests on the hemisphere at the point C , where $AC = 12a$ and angle $CAO = \alpha$, as shown in Figure 1.

Points A , C , B and O all lie in the same vertical plane.

- (a) Explain why $AO = 13a$ (1)

The normal reaction on the rod at C has magnitude kW

- (b) Show that $k = \frac{8}{13}$ (3)

The resultant force acting on the rod at A has magnitude R and acts upwards at θ° to the horizontal.

- (c) Find
- (i) an expression for R in terms of W
 - (ii) the value of θ
- (8)

6. [The centre of mass of a semicircular arc of radius r is $\frac{2r}{\pi}$ from the centre.]

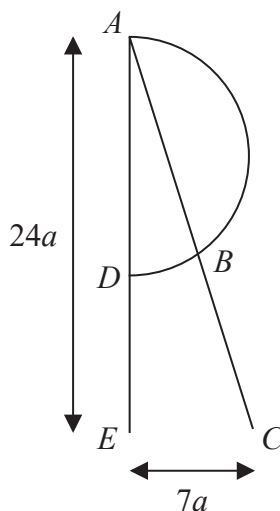


Figure 2

Uniform wire is used to form the framework shown in Figure 2.

In the framework,

- ABC is straight and has length $25a$
- ADE is straight and has length $24a$
- ABD is a semicircular arc of radius $7a$
- $EC = 7a$
- angle $AEC = 90^\circ$
- the points A, B, C, D and E all lie in the same plane

The distance of the centre of mass of the framework from AE is d .

- (a) Show that $d = \frac{53}{2(7 + \pi)}a$ (4)

The framework is freely suspended from A and hangs in equilibrium with AC at angle α° to the downward vertical.

- (b) Find the value of α . (7)

7. A particle P is projected from a fixed point O on horizontal ground. The particle is projected with speed u at an angle α above the horizontal. At the instant when the horizontal distance of P from O is x , the vertical distance of P above the ground is y . The motion of P is modelled as that of a particle moving freely under gravity.

(a) Show that $y = x \tan \alpha - \frac{gx^2}{2u^2} (1 + \tan^2 \alpha)$ (6)

A small ball is projected from the fixed point O on horizontal ground. The ball is projected with speed 20 m s^{-1} at angle θ° above the horizontal. A vertical pole AB , of height 2 m , stands on the ground with $OA = 10 \text{ m}$, as shown in Figure 3.

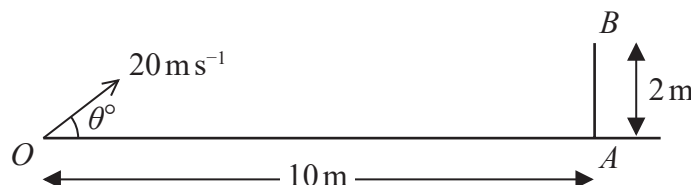


Figure 3

The ball is modelled as a particle moving freely under gravity and the pole is modelled as a rod.

The path of the ball lies in the vertical plane containing O , A and B .

Using the model,

- (b) find the range of values of θ for which the ball will pass over the pole. (3)

Given that $\theta = 40^\circ$ and that the ball first hits the ground at the point C

- (c) find the speed of the ball at the instant it passes over the pole, (5)

- (d) find the distance OC . (2)